Team members:

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Video Presenter:

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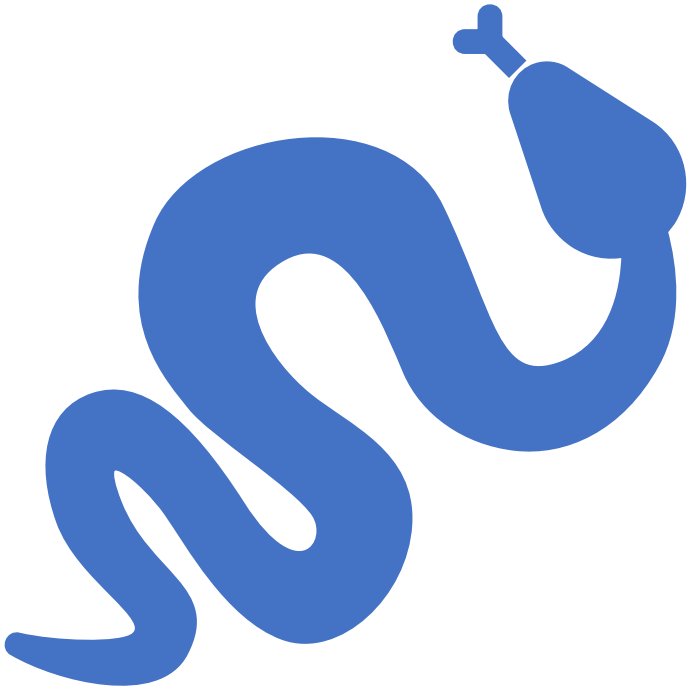
Snake Game



Outline

- Snake Pit
- Snake Initialization
- Movement & Input
- Speed Control
- Trophy System
- Game Over / Win

Team members tasks

JAKUB: SNAKE INITIALIZATION, DRAWING, GLOBAL DATA

ZACH: MOVEMENT + INPUT LOGIC

ANDREW: INPUT IMPROVEMENTS + COLLISION LOGIC

The Snake Pit

- Uses full terminal (LINES, COLS)
- Border is drawn with ncurses
- Snake is restricted inside borders

```
clear();
box(stdscr, 0, 0);
```

```
if (snake[0].x + dir_x < 1 || snake[0].x + dir_x >= COLS - 1 ||
    snake[0].y + dir_y < 1 || snake[0].y + dir_y >= LINES - 1) {
    gameover = 1;
}
```

The Initial Snake Length & Move

```
int length = 5;

int start_x = COLS / 2;
int start_y = LINES / 2;

for (int i = 0; i < length; i++) {
    snake[i].x = start_x - i;
    snake[i].y = start_y;
}
```

- Snake starts with length = 5
- Snake is spawned at center of the screen
- Initial direction is set randomly

Snake Movement

- Arrow keys + WASD supported
- Prevents reversing direction
- Snake moves by shifting body forward

```
case KEY_LEFT:
    if (dir_x != 1) {
        dir_x = -1;
        dir_y = 0;
    }
```

```
for (int i = length - 1; i > 0; i--) {
    snake[i] = snake[i - 1];
}
snake[0].x += dir_x;
snake[0].y += dir_y;
```

Snake’s Speed

- Speed increases as snake grows
- Delay decreases based on length

```
int delay = 80000 - (length * 800);
usleep(delay < 35000 ? 35000 : delay);
```

Trophy Display & Move

- One trophy is spawned at a time
- Trophies are spawned at a random position inside the border
- Avoids spawning on snake

```
for (int i = 0; i < length; i++) {
    if (snake[i].x == trophy_x && snake[i].y == trophy_y) {
        valid = 0;
    }
}
```

```
trophy_x = rand() % (COLS - 2) + 1;
trophy_y = rand() % (LINES - 2) + 1;
```

```
if ((now - trophy_spawn_time) > trophy_duration) {
    trophy_active = 0;
}
```

Trophy & Expiration

- Trophy expires after random time
- New trophy spawns automatically
- Only ONE trophy exists at a time

```
trophy_spawn_time = time(NULL);
trophy_duration = rand() % 21 + 20;
```

Trophy & Points

- Trophy value: 1-9
- Snake grows by that value
- Score increases

```
if (snake[0].x == trophy_x && snake[0].y == trophy_y) {
    score += trophy_value;
    length += trophy_value;
}
```

Snake Dies ☹

```
if (snake[0].x == snake[i].x && snake[0].y == snake[i].y) {
    gameover = 1;
}
```

Game ends when:

- Snake hits the border
- Snake hits itself
- User quits

```
mvprintw(LINES / 2, COLS / 2 - 10, "GAME OVER!");
```

Player Wins ☺

The user wins when:

- Snake's length is equal to half the perimeter

```
int win_length = (2 * (COLS + LINES)) / 2;
```

```c
        move_snake();
        trophies();
        draw();

        // Speed up slightly as length increases
        int delay = 80000 - (length * 800);
        usleep(useconds: delay < 35000 ? 35000 : delay);
    }
}



int main() {
    srand(seed: time(timer: NULL));

    initscr();            // start ncurses
    noecho();             // don't print input
    cbreak();            // disable line buffering

    keypad(stdscr, TRUE); // enable arrow keys
    nodelay(stdscr, TRUE);// non-blocking input
    curs_set(0);          // hide cursor
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
/usr/bin/ld: Snake.c:(.text+0x808): undefined reference to `LINES'
/usr/bin/ld: Snake.c:(.text+0x829): undefined reference to `mvprintw'
/usr/bin/ld: Snake.c:(.text+0x830): undefined reference to `stdscr'
/usr/bin/ld: Snake.c:(.text+0x838): undefined reference to `wrefresh'
/usr/bin/ld: Snake.c:(.text+0x847): undefined reference to `endwin'
/usr/bin/ld: warning: creating DT_TEXTREL in a PIE
collect2: error: ld returned 1 exit status
superagariogamer50@cloudshell:~$ gcc Snake.c -o snake -lncurses
superagariogamer50@cloudshell:~$ ./snake
superagariogamer50@cloudshell:~$ ./snake
superagariogamer50@cloudshell:~$ ./snake
superagariogamer50@cloudshell:~$
superagariogamer50@cloudshell:~$ ./snake
superagariogamer50@cloudshell:~$ ./snake
superagariogamer50@cloudshell:~$ ./snake
superagariogamer50@cloudshell:~$
```

d: idle Cloud Code - No Project

THANK YOU
Q/A

GitHub Repo:
https://github.com/zachsarc/SnakeGameWithC